Gustavo Furtado da Silva

Data Scientist

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Summary

Data scientist focused on end-to-end data pipelines (ingestion, ETL, data quality and orchestration) that feed machine learning models and analytics. Currently in R&D at Samsung Research Brazil; previously spent eight years at Ypê bringing pipelines, machine learning and simulation to manufacturing and sales. M.Sc. in Production Engineering and Operations Research from the Federal University of Santa Catarina. Python, SQL, Spark, Databricks, Airflow and dbt.

Professional Experience

Samsung Research Brazil, Data Science R&D Specialist — Campinas
Feb 2025 – present

- Clinical data engineering and data science for healthcare R&D projects, building data pipelines (ETL) that prepare and validate clinical data for machine learning models.
- Integrated pipelines with workflow orchestrators, focusing on data quality, lineage and reproducibility.
- Member of internal data architecture and infrastructure committees, contributing to data standards and data governance.

Ypê, Senior Big Data and Data Science Analyst — Amparo
Jun 2021 – Jan 2025

- Implemented scalable pipelines with Spark, Databricks, Airflow and dbt, reducing report generation time from one week to just a few hours and enabling real-time analytics.
- Developed machine learning models using Scikit-Learn and TensorFlow for revenue optimization, customer segmentation, product classification, and indicator forecasting, enabling faster and more accurate business decisions.
- Designed and managed a PostgreSQL database on Azure Cloud, centralizing strategic information and ensuring improved data accessibility and reliability.

Ypê, Manufacturing Data Analyst — Amparo
Oct 2018 – Jun 2021

- Developed web apps in Flask with Python and SQL Server for logistics and production data visualization and analysis, including monitoring fleet cycles and routes via GPS coordinates, optimizing times and routes.
- Supported goal management and KPIs by automating routines and training the team in visual management and daily practices.
- Participated in Industry 4.0 conferences and worked with suppliers to implement technologies such as intelligent dashboards, using simulations to validate new projects.

Ypê, Industrial Simulation Analyst — Amparo
Dec 2017 – Oct 2018

- Created process simulation models for new factories, production lines, and inventory optimization, providing data-driven insights for strategic capacity expansion decisions.
- Developed nonlinear programming models for optimizing production plans, maximizing resource usage, and reducing operational costs.
- Managed and analyzed industrial KPIs to generate strategic insights for senior management.

Wetzel S/A, Process Engineering Intern — Joinville
Mar 2011 – Mar 2013

- Conducted time studies and improved manufacturing processes.
- Developed tools to increase productivity.
- Conducted economic feasibility analyses for new processes.

Education

Federal University of Santa Catarina, Master's in Production Engineering and Operations Research — Florianópolis, Mar 2015 – Feb 2017

- Advisor: Dr. Nelson Casarotto Filho
- Thesis: Procedure for Analyzing Shared Production Systems Using Simulation-Based Optimization

State University of Santa Catarina, Bachelor's in Production Engineering and Systems — Joinville, Jan 2009 – Dec 2013

- Advisor: Dr. Evandro Bittencourt
- Final Project: Economic Feasibility Analysis for Acquiring Monitoring Software for Productive Maintenance Activities of Aluminum Injection Molding Machines

Projects

Dados Abertos Brasil

Open-source Python package, with over 190,000 downloads, that simplifies access to Brazilian government open data and APIs, such as IBGE and IPEA.

Awards

SIMEP 2017 Award for Best Operations Research Paper

Advanced English: TOEFL iBT Certified

Data Stack

Languages and Databases: Python, SQL, PySpark, Pandas, PostgreSQL, SQL Server

Data Engineering: Data Pipelines, ETL/ELT, Airflow, dbt, Spark, Databricks, Data Lakehouse, Data Quality, Data Governance

Cloud and DevOps: Azure, Docker, Git, GitHub Actions, CI/CD

Machine Learning and AI: Scikit-learn, TensorFlow, XGBoost, statsmodels, MLOps, Generative AI, LLMs

Web Apps and Dashboards: Flask, FastAPI, Plotly

Publications

Simulation-Based Analysis of Shared Manufacturing Systems — Mar 2020
Gustavo Furtado da Silva, Nelson Casarotto Filho, Enzo Morosini Frazzon
10.1590/0104-530X3718-20 (Gestão & Produção, 27(1))

Métodos de Simulação-Otimização Aplicados a Internalização de Processos Produtivos — May 2017
Gustavo Furtado da Silva, Nelson Casarotto Filho
(5th Production Engineering Symposium (SIMEP))

Simulation-Based Analysis of "One Product Integrated Manufacturing" Systems — Oct 2016
Gustavo Furtado da Silva, Enzo Morosini Frazzon, Nelson Casarotto Filho
(9th International Symposium on Industrial Engineering)

A Proposal for a Simulation-Optimization Method to Support Decision-Making in the Internalization of Production Processes — Oct 2015
Gustavo Furtado da Silva, Enzo Morosini Frazzon, Nelson Casarotto Filho
10.5585/ExactaEP.v13n3.5967 (Exacta, 13(3))

Análise de Viabilidade Financeira para a Aquisição de um Software de Monitoramento das Atividades de Manutenção Produtiva de Injetoras de Alumínio — May 2015
Gustavo Furtado da Silva, Evandro Bittencourt, Valdésio Benevenutti
(3rd Production Engineering Symposium (SIMEP))